

GENE REGULATORY NETWORK INFERENCE FROM INTEGRATIVE MULTI-OMICS DATA

By - Vishak Kashyap K, UG4 CND

Advisor - Dr Vinod PK

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Weekly Update

SC-MO-GRN-DB

- **Comprehensive Repository:** SC-MO-GRN-DB provides two things we need together: experimentally-validated "ground-truth" regulatory networks, and matched single-cell data for the same cell types.
- **Experimental Baselines:** Uses high-confidence physical evidence (ChIP-seq localization) and functional evidence (CRISPR/shRNA perturbations) rather than computational predictions.
- **Strict Cell-Type Matching:** Pairs raw, noisy sequencing data directly with exact tissue- and cell-matched reference networks to eliminate validation mismatches.
- **Unbiased Model Benchmarking:** Establishes a standardized, uniform platform to objectively test, compare, and replicate different deep learning GRN inference algorithms.
- **Integrated Graph Analytics:** Features built-in structural analysis tools to compute network centrality metrics (e.g., PageRank, degree centrality) and map TF-pathway associations.

SC-MO-GRN-DB

Filtering Single-Cell Datasets

Use the dropdown menus to filter reference networks by:

- Data Modality
- Organism
- Cell Type

Downloading Single-Cell Datasets Data

- Click "Download" next to an individual dataset to download its data in ZIP format.
- Select multiple datasets and click "Download Selected Batch" to download them as a ZIP file.

Data Modality: Organism: Cell Type:

All

All

All

Results

Select	Dataset ID	Data Modality	Organism	Cell Type	Number of Cells	PMID or DOI	Download
☐	DS001	scRNA	Mouse	ESC	422	31907445	Download
☐	DS002	scRNA	Mouse	DC	384	31907445	Download
☐	DS003	scRNA	Mouse	HSC	1072	31907445	Download
☐	DS004	scRNA	Mouse	HSC	890	31907445	Download
☐	DS005	scRNA	Mouse	HSC	848	31907445	Download
☐	DS006	scRNA	Human	ESC	759	31907445	Download
☐	DS007	scRNA	Human	HEP	426	31907445	Download
☐	DS010	scRNA/scATAC/scHiC	Mouse	ESC	9021	34579774	Download
☐	DS011	scRNA/scATAC/HiC	Mouse	ESC	36535	35977485	Download
☐	DS012	scRNA/scATAC	Mouse	ESC	930	29987051	Download
☐	DS013	scRNA/scDNAMet	Mouse	ESC	758	31827285	Download
☐	DS014	scRNA/scATAC	Mouse	ESC	68794	DOI:10.1101/2022.06.15.496239	Download
☐	DS015	scRNA/scChIP	Mouse	ESC	11875	37563440	Download
☐	DS016	scCRISPR	Human	K562	65337	27984733	Download
☐	DS017	scCRISPR	Human	K562	111445	31395745	Download
☐	DS018	scCRISPR	Human	K562	1989578	35688146	Download
☐	DS019	scRNA	Human	K562	953	26000487	Download
☐	DS020	scATAC	Human	K562	1923	28825706	Download
☐	DS021	scDNAMet	Human	K562	500	25732828	Download
☐	DS022	scHiC	Human	K562	63	38622459	Download
☐	DS023	scRNA/scHiC	Human	BJ	63	38622459	Download
☐	DS024	scRNA/scHiC	Human	GM12878	221	38622459	Download
☐	DS025	scRNA/scATAC	Human	K562	434	35501385	Download
☐	DS026	scRNA/scATAC	Human	Macrophage	3114	38747151	Download
☐	DS027	scRNA/scATAC	Human	MCF7	2995	36318267	Download
☐	DS028	scRNA	Human	GM12878	8274	31273215	Download
☐	DS029	scRNA	Human	MCF7	6256	30089904	Download
☐	DS030	scRNA	Human	HSC	183	35440586	Download
☐	DS031	scATAC	Human	BJ	96	28825706	Download
☐	DS032	scATAC	Human	H1	96	28825706	Download
☐	DS033	scATAC	Human	GM12878	384	28825706	Download

MEvD-GRN (Multi-Evidence Distillation) - Architecture

Each gene gets four input features: from scRNA-seq, the mean and variance of its expression across cells; from scATAC-seq, a gene activity score (we sum the accessibility of all open-chromatin peaks within ± 100 kb of the gene's start site, then take the mean and variance across cells). These flow through:

- RNA encoder $\rightarrow$ an RNA embedding.
- ATAC encoder $\rightarrow$ an ATAC embedding.
- Gated fusion $\rightarrow$ a learned gate decides, per gene and per feature, how much to trust RNA vs. ATAC, and blends them.
- GNN backbone (2-layer GraphSAGE) $\rightarrow$ message passing over a candidate graph so each gene absorbs context from its regulatory neighbors (which TFs could reach it, which targets a TF connects to).
- Bilinear decoder $\rightarrow$ scores a TF–target pair as $\text{sigmoid}(h_{\text{TF}} \cdot W \cdot h_{\text{target}})$. The learnable matrix W makes the score asymmetric, which matters because regulation has direction (TF A can regulate B without B regulating A).

Which datasets were used, and why?

- Our method depends on the database's three evidence tiers, so I could only use cell types that have all three. Across the entire database, exactly two cell types qualify:
 - K562 (human leukemia cell line)
 - ESC (mouse embryonic stem cells)
- For each, I paired:
 - Reference networks (three tiers each) - act as the supervision signal.
 - Single-cell RNA-seq - measures how much each gene is expressed per cell.
 - Single-cell ATAC-seq - measures which regions of DNA are "open" (accessible for a TF to bind). For K562 this came from a 10x multiome experiment; for ESC from a joint RNA+ATAC dataset.

Dataset Size for the chosen cell lines

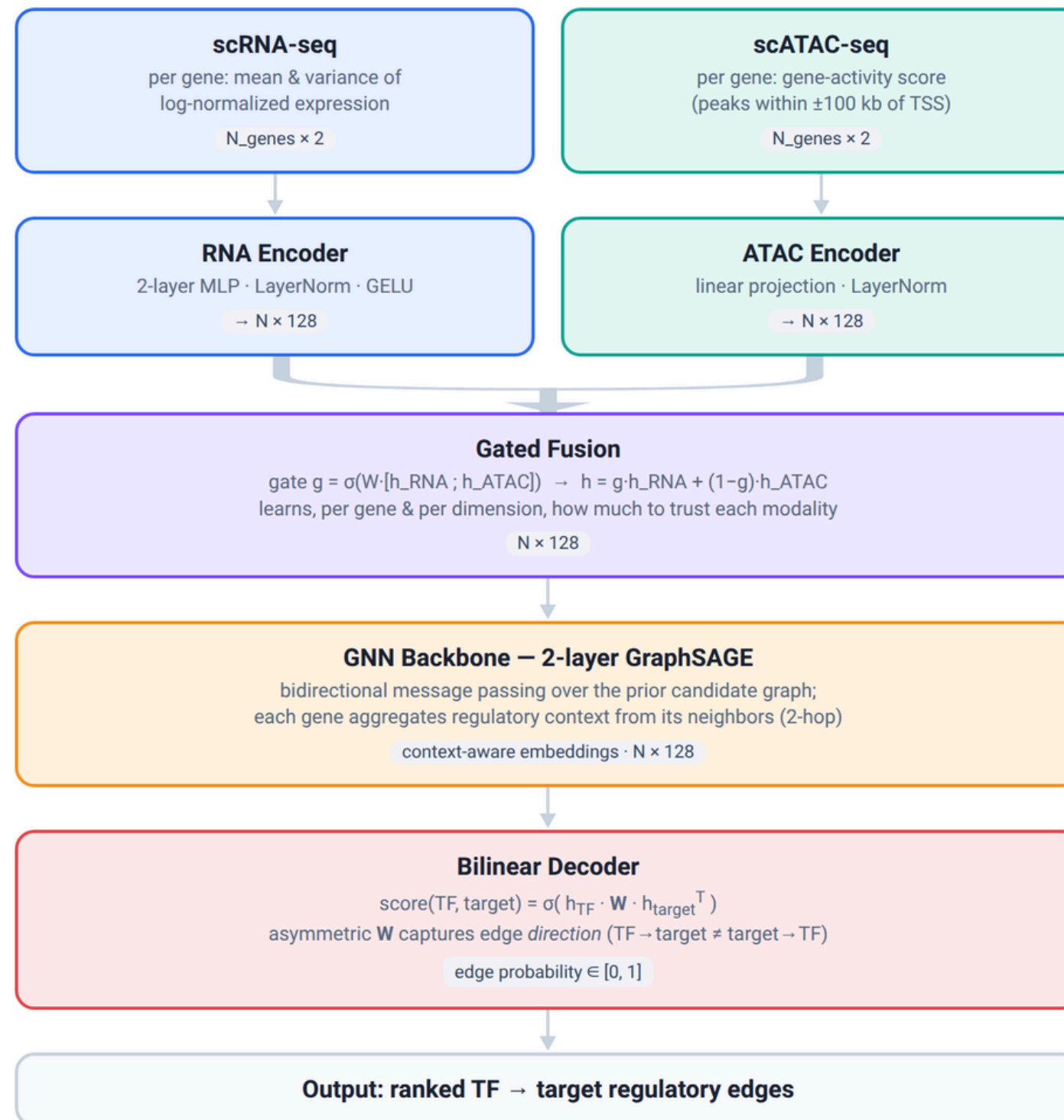
Metric	K562 (human)	ESC (mouse)
scRNA cells	~950	~8,750
scATAC cells	~1,100	~2,700
Transcription factors (TFs)	209	72
Target genes	~22,900	~14,600
Localization edges	~1.17 M	~76 K
Perturbation edges	~173 K	~69 K
Dual-evidence edges	~27 K	~7 K

How I plan on using the Reference Networks?

- The database gives every regulatory edge one of three confidence levels:
 - Localization (from ChIP-seq / ChIP-chip) - the TF is physically found bound near the target gene. Lots of edges, but binding doesn't always mean regulation.
 - Perturbation (from knockouts / CRISPR) - deleting the TF actually changes the target's expression. Fewer edges, but this is functional proof.
 - Dual-evidence - the edge has both binding and functional evidence. Fewest edges, highest confidence.
- We use these two ways:
 - as a training curriculum, going from the easy/noisy tier to the hard/high-confidence tier
 - as per-tier evaluation ground truth, so we can measure performance separately at each confidence level.

MEvD-GRN (Multi-Evidence Distillation) - Architecture

MODEL ARCHITECTURE (PER-GENE DATA FLOW)



TRAINING STRATEGY — MULTI-EVIDENCE CURRICULUM

Supervised link prediction. The database's three evidence tiers form a natural **confidence hierarchy**; we train **easy** → **hard**, fine-tuning through each tier.

Stage 1 · Localization

ChIP-seq binding — many edges, noisy signal

EASY

↓ fine-tune, lower learning rate

Stage 2 · Perturbation

knockout / CRISPR — functional & causal

HARDER

↓ fine-tune, freeze encoders

Stage 3 · Dual-Evidence

binding **AND** functional — highest confidence

HARDEST

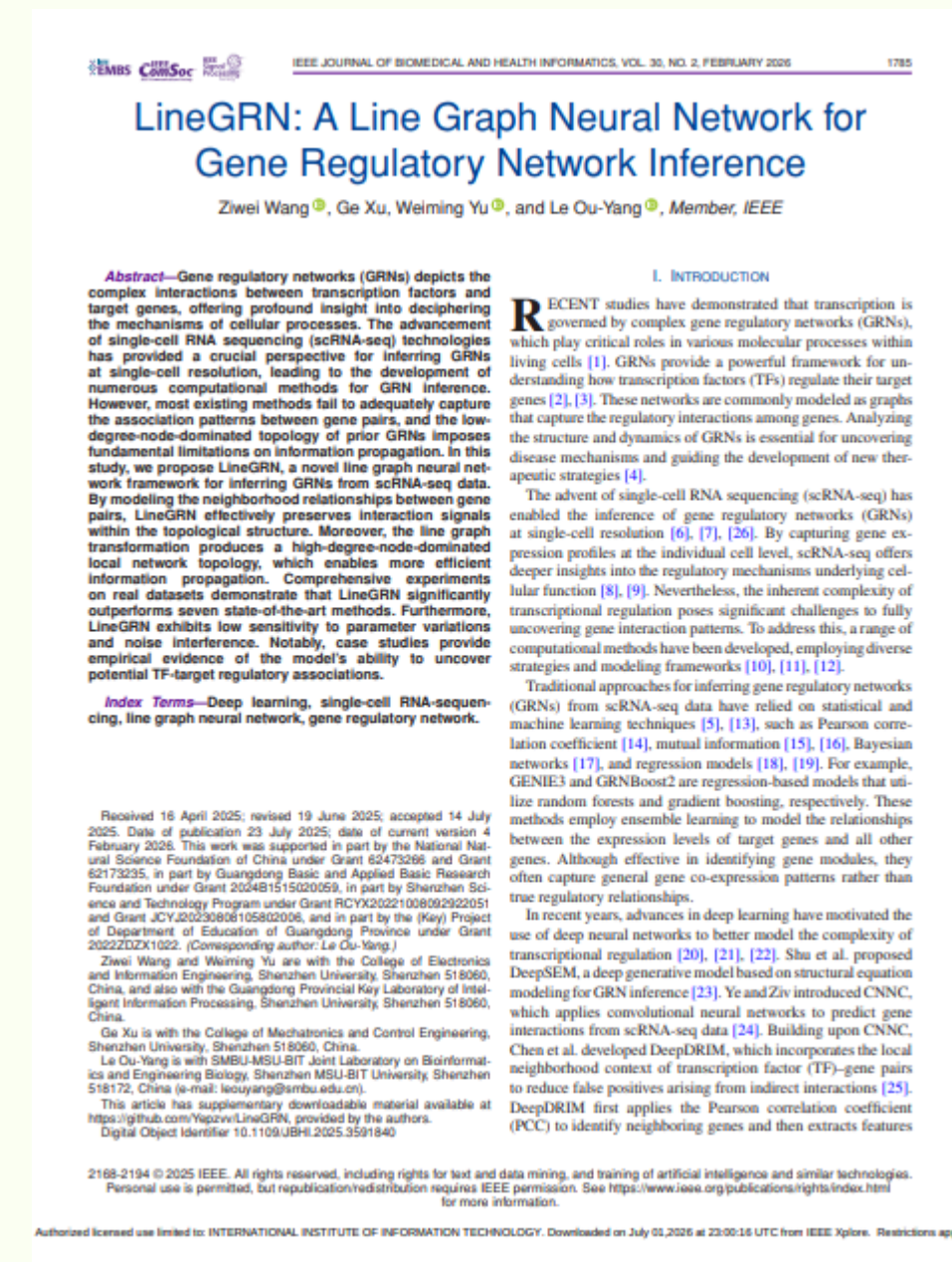
Why a curriculum? The highest-value tier (dual-evidence) is far too small to train from scratch. Earlier, larger tiers give a strong **warm start**, so the final stage only has to *specialize*. No existing GRN method uses these three tiers as a training curriculum — this is our core contribution.

Training details. AdamW + cosine-annealing LR · class-weighted BCE (focal option) · edge-level train/val/test split (no leakage) · hard negatives mined from adjacent tiers · early stopping on validation AUPR.

Evaluation. Score every candidate edge on a held-out tier, then rank. **AUPR** (primary — networks are sparse) · **AUROC** · Early Precision · EPR (enrichment over random).

Baseline Models to beat

- **Classical Statistical:**
 - GENIE3, GRNBoost2
 - PCC, PIDC
- **scRNA-only GNNs:**
 - GENELink, GNNLink
 - DeepSEM, SCODE, GNE.
- **ATAC / Motif-based:**
 - DeepTFni, SCRIP, SCENIC+
 - scMTNI, LINGER.



How Curriculum Learning is used?

- The three evidence tiers form a easy to hard ordering:
 - Stage 1 - Localization (easy): many edges, direct binding signal. The model learns the broad shape of "which TFs can reach which genes."
 - Stage 2 - Perturbation (harder): fine-tune on functional/causal edges, so the model learns to tell binding apart from actual regulation.
 - Stage 3 - Dual-evidence (hardest): fine-tune on the small, high-confidence set (both binding and functional). Here we freeze the encoders and only refine the graph/decoder, to avoid overwriting good features with signal from very few examples.
- Why do this? The most valuable tier (dual-evidence) is far too small to train a model from scratch. Curriculum learning lets the earlier, larger tiers provide a strong warm start, so by the time we reach the hard tier the model only needs to specialize, not learn from nothing.

Training, Validation, and Testing Strategy

- Ground Truth & Task Formulation: We use supervised link prediction on self-contained SC-MO-GRN-DB reference networks, treating each of the three evidence tiers as its own independent source without relying on external databases.
- Strict Edge-Level Splitting: Positive edges are split strictly at the edge level, allocating 70% Train, 15% Validation, and 15% Test independently per tier via a fixed seed to guarantee zero leakage.
- Disjoint Negative Sampling: Negative samples are random, non-regulatory TF-gene pairs drawn disjointly across splits, ensuring that a negative pair seen during training cannot reappear at test time.
- Validation for Checkpointing: The 15% validation set is used actively during training to monitor AUPR, driving early stopping and selection of the best model checkpoint.
- Held-Out Testing & Headline Metrics: The 15% test set is completely held out and evaluated only once for final reporting. We evaluate performance on all three tiers separately, but the high-confidence dual-evidence test set serves as our primary headline metric.

How AUPR and AUROC are computed?

- On a held-out test set for each tier, we combine true edges (positives) with sampled non-edges (negatives), and have the model score every candidate edge. We then rank the candidates by score:
- AUROC (Area Under the ROC Curve): the probability that a randomly chosen true edge is scored higher than a randomly chosen non-edge, an overall ranking-quality measure.
- AUPR (Area Under the Precision–Recall Curve): we sweep the score threshold, plot precision against recall, and take the area. This is our primary metric because regulatory networks are extremely sparse (far more non-edges than edges), and AUPR reflects how enriched the top predictions are for true edges. We also report Early Precision (accuracy of the top-K predictions) and EPR (how many times better than random guessing).

Preliminary Results

MEvD-GRN vs. Baselines: Dual-Evidence Tier

Cell Type	Method	AUPR x 1000	AUROC x 1000
K562	MEvD-GRN (ours)	878	976
K562	GRNBoost2	175	532
K562	GMFGRN (graph-MF)	167	530
ESC	MEvD-GRN (ours)	728	938
ESC	GMFGRN (graph-MF)	177	523
ESC	GRNBoost2	171	529

Applying the K562-trained model to ESC (with zero retraining) yields a dual-evidence AUROC of 0.77.

MEvD-GRN Across All Three Evidence Tiers
(Values formatted as: AUPR / AUROC)

Cell Type	Localization	Perturbation	Dual-evidence
K562	0.71 / 0.64	0.34 / 0.71	0.88 / 0.98
ESC	0.45 / 0.80	0.36 / 0.67	0.73 / 0.94

Ablation
Studies
(Dataset: K562,
Metric: Dual-
evidence AUPR)

Variant	AUPR x 1000	Contribution / Takeaway
Full curriculum (ours)	890	Baseline for comparison
- ATAC (RNA only)	740	Multimodal input adds +0.15
No curriculum (all tiers at once)	709	Curriculum ordering adds
No GNN	634	Graph message-passing is critical
Localization only	638	Single tier alone is far worse
Dual-evidence only	623	Cannot learn from the small tier
Perturbation only	165	Collapses without Stage-1 warm start



THANK YOU!

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